

llmXive Automated Review of On the Geometry of On-Policy Distillation

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper makes several specific quantitative claims regarding parameter-space diagnostics

(update norms, principal angles, stable ranks) that are currently not fully supported by the provided LaTeX source text. While the figures (e.g., `figures/intrinsic_metrics_3panel.pdf`, `figures/fig1_spectral_geometry.pdf`) likely contain these values, the text itself frequently cites specific numbers (e.g., “0.452”, “10°”, “1°”) without defining them in the surrounding prose or captions.

Specifically, in Section 5, the claim that OPD’s cumulative update norm is “substantially larger” than RLVR’s is supported by specific values (0.452 vs 0.168) that do not appear in the provided text. Similarly, in Section 3, the claim that SFT induces angles “above 10°” while OPD is “around 1°” lacks the specific data points in the text body. To ensure claim accuracy, these numerical values must be explicitly stated in the text or figure captions so that the reader can verify the “relaxed off-principal” and “subspace locking” assertions without relying solely on visual inspection of the figures.

Additionally, in Section 6, the transition point where objective mixing breaks the “locked regime” is described qualitatively (“when the OPD component becomes weak”). The text should specify the exact alpha threshold (e.g., $\alpha < 0.25$) observed in the data to substantiate the claim of a distinct spectral regime boundary. Without these specific data points in the text, the strength of the claims regarding the magnitude of differences between SFT, OPD, and RLVR is not fully verifiable from the manuscript text alone.

Action items.

- **[writing]** In Section 5 (Subspace Locking), the text claims OPD has a ‘substantially larger cumulative update norm than RLVR’ and cites specific values (0.452 vs 0.168). These specific numbers are absent from the provided LaTeX source and figures. Verify these values are present in the rendered PDF (Figure 5b) or remove the specific numbers from the text to avoid unsupported claims.
- **[writing]** In Section 3 (Locating OPD), the text states SFT principal angles ‘rise above 10°’ and OPD angles are ‘around 1°’. These specific quantitative claims are not visible in the provided text source. Ensure these values are explicitly reported in the figure captions or the main text to support the ‘relaxed off-principal’ claim.
- **[writing]** In Section 6 (Controls), the text claims ‘OPD-dominant mixtures retain the OPD-like stable-rank trajectory’ while ‘weak’ OPD components ‘enter a distinct spectral regime’. The specific threshold for ‘weak’ (e.g., $\alpha < 0.25$) is not defined in the text. Clarify the quantitative boundary where the trajectory departs to support the claim of a ‘boundary of the locked regime’.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

Figure 1 effectively communicates the conceptual geometry of OPD through clear visual metaphors and a consistent color scheme. However, the quantitative inset in panel (b) lacks a clear x-axis definition in the legend, and the axis label formatting is slightly inconsistent.

Figure 2

Figure 2 presents clear comparative plots of spectral geometry across methods, but key axis labels ('k', 'NSS', 'Weight matrix index') lack definitions in the caption, reducing interpretability for readers unfamiliar with the notation.

Figure 3

Figure 3 effectively visualizes update-mask localization with clear heatmaps and a colorbar, but the 'Global' row plots lack explicit y-axis labels and error bar definitions, hindering full interpretation of the quantitative claims.

Figure 4

The figure effectively visualizes the three metrics described in the caption, but the x-axis label is ambiguous regarding the training unit, and the 'OPD band' annotation in panel (a) lacks a definition in the text.

Figure 5

The figure is visually clear and effectively demonstrates the claim that OPD locks onto its final update channel earlier than SFT and RLVR. However, the caption is unrendered LaTeX code, and the y-axis notation lacks a precise definition in the text.

Figure 6

The figure's visual data contradicts the caption's claim that OPD remains intact while SFT degrades under rank-16 projection. Additionally, the legend labels 'plain' and 'K16' are undefined in the caption, obscuring method identification.

Figure 7

The figure displays subspace similarity trends but fails to visually demonstrate the 'locking' behavior described in the caption, and the y-axis label lacks the specificity found in the caption text.

Figure 8

The figure contains a contradiction between the caption’s claim that OPD is ‘intact’ and the visual data showing a performance drop for the ‘plain’ method at 100% training. Additionally, panel (b) lacks a legend to define its series.

Figure 9

The figure effectively visualizes the stability of the OPD trajectory under different conditions, but the mapping between the legend labels (Top-KL, Random) and the caption’s categories (Runtime perturbations) is implicit rather than explicit, and the use of ‘a’ instead of *lpha* in the legend is a minor notation inconsistency.

Figure 10

The figure effectively visualizes the performance of OPD versus SFT under rank constraints across benchmarks, but the legend labels (‘plain’, ‘K16’) do not explicitly match the caption’s terminology (‘unconstrained’, ‘rank-16 projected’), and the y-axis metric definitions are abbreviated.

Figure 11

The figure displays the intended metrics clearly, but the caption contains a broken cross-reference to a missing figure number, and the variable ‘a’ used in the legend is not defined in the text.

Action items.

- **[science]** Figure 1: The inset line plot in panel (b) shows ‘Stable Rank’ on the y-axis, but the legend defines the lines as SFT, OPD, and RLVR without specifying the x-axis variable (e.g., training steps or epochs), making the ‘rapidly enters’ claim in the caption difficult to verify quantitatively.
- **[writing]** Figure 1: The x-axis label in the inset plot of panel (b) reads ‘Training Progress (%)’, but the tick marks (20, 40, 60, 80, 100) lack a ‘%’ symbol, creating a minor inconsistency between the label and the axis values.
- **[writing]** Figure 2: The x-axis label ‘Rank index k’ is used in the left two columns, but the caption does not define ‘k’ (e.g., as rank index, layer index, or training iteration), creating ambiguity.
- **[writing]** Figure 2: The y-axis label ‘NSS’ in the rightmost column is undefined in the caption; the acronym should be spelled out or explained.
- **[writing]** Figure 2: The ‘Max Principal Angle (all-layer)’ plots use ‘Weight matrix index’ on the x-axis, but the caption does not clarify what this index represents (e.g., layer number, parameter block).
- **[science]** Figure 3: The ‘Global’ row contains three line plots with different y-axis scales (0-100, 0-50, 0-100) but no explicit axis labels or units are provided on the plots themselves.

making it impossible to verify the ‘Global’ metric values against the heatmap percentages without guessing.

- **[writing]** Figure 3: The ‘Global’ row plots lack y-axis labels; while the caption mentions ‘update-mask localization’, the specific metric being plotted (e.g., ‘Overlap %’, ‘Density’) is not explicitly labeled on the axes, relying on the reader to infer from the heatmaps above.
- **[science]** Figure 3: The ‘Global’ row plots show single points with error bars (or shaded regions) but do not specify what the error bars represent (e.g., std dev, confidence interval) in the caption or on the plot, which is critical for interpreting the ‘robustness’ claim.
- **[writing]** Figure 4: The x-axis label ‘Training progress (%)’ is ambiguous; the caption refers to tracking cumulative updates W_t over time, but it is unclear if the x-axis represents training steps, epochs, or percentage of total budget.
- **[writing]** Figure 4: Panel (a) contains a text annotation ‘OPD band’ pointing to the blue line, but this specific term is not defined in the caption or legend, making the annotation’s purpose unclear.
- **[writing]** Figure 5: The provided caption is a raw LaTeX comment block (starting with %) rather than a rendered description, making it difficult to read and unprofessional.
- **[writing]** Figure 5: The y-axis label ‘Sim₁₆(t, t_{final})’ uses subscript notation that is not explicitly defined in the caption, which only vaguely mentions ‘Top-16 subspace similarity’.
- **[science]** Figure 6: The caption claims ‘Rank-16 projection leaves OPD intact but degrades SFT’, but the plot shows ‘plain’ (OPD) performance dropping significantly after the projection start (dashed line), while ‘K16’ (SFT) performance improves. The visual data directly contradicts the caption’s claim.
- **[writing]** Figure 6: The legend labels ‘plain’ and ‘K16’ are not defined in the caption, making it impossible to identify which method corresponds to OPD or SFT without external context.
- **[science]** Figure 7: The caption claims OPD ‘locks onto its final update channel earlier’ than SFT and RLVR, but the plot shows OPD starting with higher similarity and rising monotonically without a distinct ‘lock’ or plateau phase that would demonstrate early locking compared to the other methods.
- **[writing]** Figure 7: The y-axis label ‘Sim(, _final)’ is ambiguous; the caption specifies ‘Top-16 subspace similarity’, but the axis label does not explicitly state this metric, potentially confusing readers about whether it is parameter similarity or subspace similarity.
- **[science]** Figure 8: The caption claims ‘Rank-16 projection leaves OPD intact but degrades SFT,’ but the plot shows the ‘plain’ (OPD) line dropping significantly at 100% training, while the ‘K16’ (projection) line remains stable. The visual data contradicts the claim that OPD is ‘intact’ at the end of training.
- **[writing]** Figure 8: The legend in panel (a) defines ‘plain’, ‘K16’, and ‘shared resume’, but panel (b) lacks a legend, forcing the reader to assume the same mapping applies without explicit confirmation.

- **[science]** Figure 9: The caption claims the figure shows ‘Runtime perturbations’ and ‘objective-level interpolation’, but the legend labels these as ‘Top-KL’, ‘Random’, and ‘Objective mixing’ (alpha values). The connection between the legend terms and the caption’s conceptual categories is not explicit, making it difficult to verify the claim that ‘Runtime perturbations’ (likely Top-KL/Random) preserve the trajectory without external inference.
- **[writing]** Figure 9: The legend in panel (c) uses the symbol ‘a’ for the parameter alpha (e.g., ‘a=0.25’), which is non-standard and potentially confusing compared to the Greek letter *lpha* used in the caption text.
- **[science]** Figure 10: The legend defines ‘K16’ (orange squares) and ‘plain’ (blue circles), but the caption claims to compare ‘unconstrained training’ and ‘rank-16 projected training’. The legend fails to explicitly label the orange line as the ‘rank-16 projected’ condition, creating ambiguity between the legend text and the caption’s description.
- **[writing]** Figure 10: The y-axis labels use abbreviated metrics (e.g., ‘avg@32 (%)’, ‘avg@4 (%)’) without defining what ‘avg’ refers to (e.g., average accuracy, average score) in the caption or axis text.
- **[writing]** Figure 11 caption contains a broken cross-reference: ‘analyzed in Figure .’ is missing the figure number.
- **[science]** Figure 11: The legend in the bottom-left panel defines ‘a=0.75’ through ‘a=0.01’ and ‘a=0.01’ (red dashed), but the caption does not define what ‘a’ represents (e.g., interpolation coefficient), making the ‘objective interpolation’ claim ambiguous.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The manuscript relies heavily on specialized terminology from linear algebra and optimization theory, often introducing these terms without sufficient scaffolding for a broader audience. While the target audience for this venue likely understands these concepts, the paper’s claim to characterize “geometry” suggests a need for clarity on the geometric terms themselves.

Specifically, the term “LRM” (Large Reasoning Model) is used in the very first sentence of the Introduction without being spelled out. While common in recent literature, strict adherence to plain language requires defining the acronym at first use. Similarly, “ULP” (Unit in the Last Place) appears in Equation 9 and the Appendix to define the bf16 realization gate. This is a low-level numerical analysis term that is not standard in high-level ML discourse; its usage here is precise but opaque to readers who do not specialize in hardware-aware training.

The concept of “stable rank” is central to the “subspace locking” argument (Section 5.1). The authors provide the formula but do not explicitly define the term’s intuitive meaning

(effective dimensionality) in the text, relying on the reader to infer it from the equation or prior knowledge. A brief explanatory clause would significantly improve accessibility. Finally, “MoE” (Mixture-of-Experts) is used in Table 1 and the text to describe a teacher variant. While a standard acronym in the field, it is not defined in the main body, creating a minor barrier for readers whose expertise is strictly in reinforcement learning or distillation rather than model architecture.

These instances represent a pattern of assuming domain-specific fluency where a brief definition would suffice to make the “geometry” arguments accessible to a wider scientific audience.

Action items.

- **[writing]** The manuscript relies heavily on specialized terminology from linear algebra and optimization theory, often introducing these terms without sufficient scaffolding for a broader audience. While the target audience for this venue likely understands these concepts, the paper’s claim to characterize “geometry” suggests a need for clarity on the geometric terms themselves. Specifically, the term “LRM” (Large Reasoning Model) is used in the very first sentence of the Introduction without being spelled

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper presents a coherent logical framework linking the “relaxed off-principal regime” to “subspace locking” and finally to “objective composition” as the controlling factor. The progression from static diagnostics (Section 4) to trajectory analysis (Section 5) and control experiments (Section 6) is logically sound. The conclusion that OPD is distinct from a simple interpolation between SFT and RLVR follows well from the presented evidence of early subspace emergence and robustness to runtime perturbations.

However, there are two specific logical gaps regarding the mathematical implementation of the claims:

1. **Projection Dimensionality (Section 5.2):** The paper claims that constraining training to a rank-16 subspace preserves OPD performance. Equation 9 defines the projection as $g \leftarrow gV_{16}V_{16}^T$, where V_{16} are the right singular vectors of the cumulative update ΔW . For a weight matrix $W \in \mathbb{R}^{d_{out} \times d_{in}}$, the right singular vectors correspond to the input dimension (d_{in}). The text must explicitly clarify whether the gradient g is being projected on its input or output dimension. If g is projected on the wrong dimension (e.g., projecting rows when V spans columns), the “bottleneck” logic fails. The current text assumes the reader infers the correct matrix multiplication order, which is a logical ambiguity in a technical proof.
2. **Causal Mechanism of Objective Mixing (Section 6.2):** The paper argues that objective mixing breaks the lock because it changes the gradient source, whereas runtime

perturbations do not. The mechanistic explanation (Eq. 13-14) posits that the covariance Σ_α is a weighted sum of Σ_{OPD} and Σ_{RLVR} . The logical leap is that this mixture *immediately* shifts the dominant subspace away from the OPD lock. The paper does not address the potential for the cross-covariance term Σ_{cross} to maintain the OPD geometry at intermediate mixing ratios (e.g., $\alpha = 0.5$). Without evidence that the cross-term is negligible or that the eigenvalues of Σ_{OPD} and Σ_{RLVR} are sufficiently separated to cause an immediate switch, the claim that the “lock” is broken solely by objective composition is not fully supported by the provided derivation.

These issues require clarification in the text or additional analysis to ensure the causal claims are rigorously supported.

Action items.

- **[science]** In Section 5.2, Eq. 9 projects gradients using right singular vectors V_{16} . The text must explicitly confirm this projects the correct dimension (input vs output) to logically enforce the claimed rank-16 bottleneck on the update subspace.
- **[science]** In Section 6.2, the claim that objective mixing breaks the lock assumes the cross-covariance term in Eq. 14 is negligible. The paper lacks evidence that this term does not maintain OPD geometry at intermediate mixing ratios, weakening the causal claim.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper makes several strong claims regarding the distinctiveness of On-Policy Distillation (OPD) that slightly overreach the empirical evidence provided.

First, the abstract and introduction repeatedly characterize OPD as “not merely an intermediate point” but as inducing its “own update geometry” and occupying a “distinct parameter-space regime” (Abstract; Introduction). However, the data in Section 4 (Table 1, Figure 1) consistently places OPD *between* SFT and RLVR on every metric: update sparsity (51.6% vs 8.1% and 77.2%), principal-angle rotation ($\sim 1^\circ$ vs $> 10^\circ$ and $< 0.5^\circ$), and spectral drift. The authors define this as a “relaxed off-principal regime,” which is effectively a description of an intermediate position with specific properties. Claiming it is a “distinct regime” rather than a specific point on the SFT-RLVR continuum risks over-interpreting a gradient difference as a categorical one. The language should be adjusted to emphasize that OPD occupies a *specific, stable region* within the spectrum, rather than implying it exists outside the interpolation logic.

Second, the claim of “functional sufficiency” for the locked subspace (Abstract; Section 5) is slightly too strong. The experiment in Figure 5 demonstrates that constraining training to the early rank-16 subspace *preserves* performance compared to unconstrained training. While this shows the subspace is *sufficient* to achieve the result, it does not prove that the full-rank

trajectory is unnecessary or that the subspace is the *unique* functional manifold. The phrasing “functionally sufficient” implies a completeness that the ablation does not strictly prove (i.e., it doesn’t rule out that the full trajectory might offer better convergence speed or robustness to distribution shift, even if final accuracy is similar). The text should clarify that the early subspace is sufficient to *maintain* the observed performance, rather than being the sole functional driver.

Finally, the “Relaxed Three-Gate Account” in Section 4.3 presents a theoretical derivation (Eqs. 10-14) as a definitive explanation for the observed geometry. While the authors correctly note in the Limitations that these are “mechanistic explanations consistent with the evidence, rather than complete causal... theories,” the main text uses definitive causal language (e.g., “This yields larger effective updates,” “This explains why...”). Given that the derivation relies on assumptions (e.g., local quadratic budget, specific covariance structures) that are not empirically verified in the paper, the claims should be softened to reflect that this is a *plausible* mechanistic hypothesis consistent with the data, rather than a proven causal account.

Action items.

- **[writing]** The claim that OPD is ‘not merely an intermediate point’ (Abstract, Intro) overstates the evidence. The data shows OPD occupies a position *between* SFT and RLVR on all metrics (sparsity, rotation, drift). The distinction is one of degree (a ‘relaxed’ regime), not a fundamentally different categorical state. The text should be tempered to reflect that OPD is a distinct *point* on the continuum, not a separate regime.
- **[writing]** The assertion that the locked subspace is ‘functionally sufficient’ (Abstract, Sec 5) extrapolates beyond the provided data. The experiment shows performance is *preserved* under a rank-16 constraint, but this does not prove the subspace is the *only* sufficient path or that the full-rank trajectory adds no value. The claim should be qualified to state that the early subspace is ‘sufficient to maintain performance’ rather than ‘functionally sufficient’ in a general sense.
- **[writing]** The mechanistic explanation in Section 4.3 (‘A Relaxed Three-Gate Account’) presents a theoretical derivation as a definitive explanation for the observed geometry. The paper admits in the Limitations that these are ‘mechanistic explanations consistent with the evidence, rather than complete causal... theories.’ The main text should avoid definitive causal language (e.g., ‘This yields...’, ‘This explains...’) and instead frame these as plausible hypotheses consistent with the data.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The manuscript demonstrates strong adherence to safety and ethical standards appropriate

for a theoretical analysis of LLM training dynamics. The authors explicitly disclose the use of generative AI (ChatGPT) for writing assistance, LaTeX formatting, and script drafting in the “AI Usage” section (Appendix), while clearly delineating that no AI system was used to generate experimental results or make autonomous scientific decisions. This transparency mitigates concerns regarding undisclosed AI authorship or data fabrication.

Regarding data privacy and consent, the study utilizes publicly available model checkpoints (Qwen3 series) and open-source datasets (DAPO-Math-17k, Dolci-Think SFT, DeepCoder, Live-CodeBench). The “Artifact Use” section confirms that these resources were used in accordance with their intended research purposes and that the authors verified public licenses or usage terms where available. No human subjects, private user data, or sensitive information were involved in the experiments, eliminating risks related to IRB/IACUC approval or informed consent.

The paper does not present immediate dual-use risks. The research focuses on characterizing the geometric properties of weight updates during on-policy distillation. While the findings could theoretically inform more efficient model training, the analysis is diagnostic and descriptive rather than prescriptive for generating harmful capabilities. The “Limitations” section appropriately contextualizes the scope of the findings to specific model families and reasoning tasks, preventing overgeneralization. No conflicts of interest are apparent in the provided text. The manuscript is ethically sound for publication.

Scientific Evidence

Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling empirical characterization of On-Policy Distillation (OPD) update geometry, supported by a suite of parameter-space diagnostics (update sparsity, principal-angle rotation, spectral drift). The evidence for OPD occupying a “relaxed off-principal regime” is robust, with consistent results across multiple model scales, teacher configurations, and data domains (Table 1, Fig 1). The use of bf16-aware sparsity is a methodological strength, addressing the quantization artifacts common in large model analysis.

However, the evidence for the central claim of “subspace locking” (Sec 5) requires further validation to rule out alternative explanations. The claim that OPD “rapidly enters” a low-dimensional channel relies on subspace similarity metrics computed at relatively sparse checkpoints (e.g., steps 63, 127, 191). Without finer-grained trajectory data, it is difficult to distinguish a true “rapid lock” from a gradual convergence that happens to appear early due to the sampling interval. Furthermore, the functional sufficiency experiment (rank-16 projection) uses a single fixed dimension ($K=16$). While justified as being near the observed stable rank, the lack of a sensitivity analysis (e.g., varying K) leaves open the possibility that the observed robustness is specific to that exact dimension rather than a general property of the subspace.

Finally, the control experiments in Section 6, particularly the objective mixing, assume that

linear interpolation of advantage signals isolates the objective composition without introducing confounding changes in gradient scale or optimization dynamics. The paper should clarify if gradient norms were monitored or normalized in these mixed-objective runs to ensure the observed spectral shifts are purely directional. Addressing these points would significantly strengthen the causal claims regarding the mechanism of subspace locking.

Action items.

- **[science]** The claim that OPD ‘rapidly enters’ a locked subspace (Sec 5.2) relies on subspace similarity metrics computed at discrete checkpoints (63, 127, 191 steps). The evidence does not rule out that the lock emerges between these checkpoints or that the ‘early’ alignment is an artifact of the specific 20% projection start point. Provide a finer-grained trajectory analysis or sensitivity analysis of the projection start time to confirm the ‘rapid’ nature of the lock.
- **[science]** The functional sufficiency claim (Sec 5.2) is based on a single projection dimension ($K=16$). While the text notes this is ‘close to’ the observed stable rank, the evidence does not demonstrate that the result is robust to small variations in K (e.g., $K=12$ or $K=20$). A sensitivity analysis of the rank-constrained training results is required to ensure the ‘sufficiency’ is not an artifact of the specific $K=16$ choice.
- **[science]** The control experiment mixing OPD and RLVR objectives (Sec 6.2) uses a linear interpolation of advantage signals. The claim that this isolates ‘objective composition’ as the driver assumes the mixing does not introduce confounding variance in gradient magnitudes or optimization dynamics. The paper should report whether the mixed objectives were normalized or if gradient norms were controlled to ensure the spectral shift is due to direction, not scale.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The statistical analysis of parameter-space geometry is conceptually sound, utilizing established diagnostics like stable rank, principal angles, and spectral drift. However, the presentation of results lacks necessary statistical rigor to fully support the strong claims of distinct regimes.

First, the paper reports point estimates for key metrics (e.g., “principal angles rise above 10° ” for SFT vs “around 1° ” for OPD in Section 4.2) without providing measures of uncertainty. Given that these metrics are averaged across layers and potentially across seeds, the absence of standard deviations, confidence intervals, or error bars in the text (and implied in the figures) makes it difficult to assess the statistical significance of the observed differences. The claim that OPD occupies a “distinct” regime relies on these differences being robust, not just point-estimate separations.

Second, the definition of “bf16-aware update sparsity” depends critically on the threshold $\eta = 10^{-3}$ (Section 4.1). The paper does not include a sensitivity analysis to show how the reported sparsity values (e.g., 51.6% for OPD) change if η is varied slightly. Without this, the conclusion that OPD is “intermediate” could be sensitive to the arbitrary choice of precision cutoff.

Finally, the functional sufficiency experiment (Section 5.2) claims that rank-16 projection “degrades SFT” while leaving OPD “essentially unchanged.” While the performance drop is described, the statistical significance of this difference is not quantified. Were the evaluations run with multiple seeds? Is the difference in final accuracy statistically significant? Adding error bars or reporting p-values for these comparative experiments would strengthen the evidence for the “subspace locking” hypothesis.

Action items.

- **[science]** The paper reports specific quantitative values for stable rank, Frobenius norms, and principal angles (e.g., ‘10 degrees’, ‘ 10^{-3} level’) in Section 5 and Figure 5 captions. However, no standard deviations, confidence intervals, or p-values are provided to assess the statistical significance of the differences between SFT, OPD, and RLVR trajectories. Given the claim of distinct regimes, variance estimates across seeds or layers are required to rule out noise.
- **[science]** The ‘bf16-aware update sparsity’ metric relies on a fixed threshold $\eta=10^{-3}$ (Section 4.1). The sensitivity of the reported sparsity percentages (e.g., 51.6% vs 77.2%) to this specific threshold choice is not analyzed. A sensitivity analysis varying η is needed to ensure the ‘relaxed off-principal’ conclusion is not an artifact of the precision cutoff.
- **[science]** In the rank-16 projection experiment (Section 5.2), the performance difference between OPD and SFT is presented as a definitive finding. However, the text does not mention whether the evaluation was repeated across multiple random seeds or if statistical tests (e.g., t-tests) were performed to confirm the degradation in SFT is significant compared to the variance in OPD.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The manuscript demonstrates a high standard of academic writing, with a clear, logical flow and precise technical vocabulary appropriate for the field. The abstract effectively summarizes the core contributions, and the introduction successfully frames the research questions. The prose is generally concise, and the argumentation is well-structured, guiding the reader from the problem statement through the methodology to the conclusions.

However, there are a few minor typographical and stylistic issues that detract slightly from

the overall polish. In the Introduction (Section 1), there is a noticeable double space in the phrase “with a bias toward the RLVR side.” While minor, such errors should be eliminated in a final submission. Additionally, some sentences in the Introduction and Section 3 could be slightly tightened for maximum clarity. For instance, the explanation of why OPD’s trajectory is difficult to infer could be made more explicit to ensure the reader immediately grasps the nuance of the hybrid nature of the method.

The use of the “takeawaybox” environment is effective for highlighting key implications, and the transitions between sections are smooth. The mathematical notation is consistent and well-integrated into the text. Overall, the writing quality is strong, and the identified issues are easily rectifiable with a careful proofread.

Action items.

- **[writing]** In Section 1 (Introduction), the phrase ‘with a bias toward the RLVR side’ contains a double space between ‘a’ and ‘bias’. This is a minor typographical error that should be corrected for professional polish.
- **[writing]** In Section 1, the sentence ‘...dense token-level distillation resembles SFT, while on-policy sampling and policy-gradient optimization connect it to RLVR’ is slightly ambiguous. Consider clarifying that the *combination* of these features makes the trajectory difficult to infer, or rephrase to ensure the causal link is explicit.
- **[writing]** In Section 3 (Locating OPD), the phrase ‘...with a bias toward the RLVR side’ appears again. Ensure consistency in terminology and check for any remaining spacing issues in the final proof.